Academic Report: Machine Learning for Clad Geometry Prediction in Laser Metal Deposition (LMD)

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1. Introduction and Research Context

This study investigates the application of machine learning (ML) to predict clad geometry in Laser Metal Deposition (LMD) — a directed energy deposition (DED) additive manufacturing process. The key challenge is the inherently **small experimental dataset** (n = 89 samples), which places strict constraints on model complexity and demands rigorous validation strategies. The study progressed systematically from exploratory analysis through baseline linear regression to advanced nonlinear and ensemble models, with the goal of establishing robust, generalizable predictors for critical LMD output targets including **clad height**, **aspect ratio**, and **porosity**. This modelling pipeline provides foundational contributions to data-driven process control for metal additive manufacturing.

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2. Dataset Overview & Preprocessing

2.1 Feature Space and Target Variables

The processed dataset contains **89 observations** and **3 input process parameters** as features: **Power (W)**, **Scanning Speed (mm/s)**, and **Feed Rate (g/min)**. The deliberate exclusion of derived energy metrics (e.g., LED, VED) was motivated by their near-perfect linear dependence on the primary inputs, which would have introduced severe multicollinearity.

The target variables investigated include:

- **Clad height** (µm) — mean ~803 µm, std ~429 µm
- **Aspect ratio** — a geometric ratio of bead geometry
- **Porosity** (%) — mean ~0.30, std ~0.47, with a highly right-skewed distribution (max = 3.285%)

The 80/20 train-test split yielded **71 training and 18 test samples**, consistent with small-data best practices.

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3. Exploratory Data Analysis (EDA)

3.1 Distribution Analysis

Histograms and summary statistics revealed that **porosity is heavily right-skewed** (near-zero mode), making it inherently difficult to model with linear assumptions. The clad height distribution is unimodal but wide-ranging (214–2432 µm), suggesting strong process sensitivity to input parameters.

3.2 Multicollinearity (VIF Analysis)

A Variance Inflation Factor (VIF) analysis was conducted to assess multicollinearity among the three input features:

Feature	VIF
Power	14.23
Speed	10.48
Feed	5.68

Power and Speed exhibit **severe multicollinearity** (VIF > 10), which inflates standard errors of regression coefficients and destabilizes linear model estimates. Feed is at an acceptable level (VIF < 10). This finding was critical in the decision to avoid regularization-free OLS for production deployment and to monitor coefficient interpretations carefully.

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4. Baseline Linear Regression Study

4.1 Model Framework

A full ML pipeline was built using **scikit-learn** with `StandardScaler` preprocessing within a 5-fold cross-validation framework (`KFold, shuffle=True, random_state=42`). Models evaluated in the comparison stage included: **OLS Linear Regression**, **Ridge** (α=1.0), **Lasso** (α=0.1), **RidgeCV**, and **LassoCV**.

4.2 Height Target — Linear Regression Performance

For clad height (`heightum`), the linear regression model yielded satisfactory in-sample and cross-validated performance:

Metric	Train	Test	CV Mean
R^2	0.783	0.783	0.755
RMSE	209.64 µm	169.67 µm	202.18 µm
MSE	43948.9	28787.6	40877.4
MAE	144.39 µm	136.85 µm	—

The OLS summary (via `statsmodels`) confirmed **statistical significance** of all coefficients ($p < 0.001$), an F-statistic of 115.4 ($p < 0.001$), and an Adjusted $R^2 = 0.766$. The **Durbin-Watson**

statistic of 2.289 indicated no significant autocorrelation in residuals — a healthy diagnostic. However, the **Omnibus test** (p = 0.000), **Skewness = 1.874**, **Kurtosis = 8.288**, and **Jarque-Bera = 124.257** collectively pointed to **non-normally distributed residuals**, violating a core OLS assumption and suggesting that a linear model is insufficient to capture all data variance.

4.3 Porosity Target — Linear Regression Breakdown

For **porosity**, the linear regression model failed almost completely:

Model	R^2_train	R^2_test	RMSE_CV
Linear Regression	0.434	−5.134	0.363
Ridge	0.434	−5.061	0.362
Lasso	0.331	−3.266	0.397
RidgeCV	0.429	−4.518	0.358
LassoCV	0.433	−4.849	0.363

The severely negative test R^2 values indicate that all linear models **overfit the training data** and fail to generalize. CV R^2 values hover around −0.64 to −0.43, confirming that linear relationships between Power/Speed/Feed and porosity are weak. This is consistent with the physics of LMD: porosity is governed by highly nonlinear melt pool thermodynamics and solidification behavior that are not well captured by linear regression.

> **Key Learning:** The consistent failure of regularized linear models on porosity reveals the physical nonlinearity of this target and motivates the adoption of nonlinear and ensemble approaches.

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5. Nonlinear and Ensemble Model Study

5.1 Model Suite and Hyperparameter Tuning

A comprehensive nonlinear study was conducted for both **height** and **porosity** targets using `GridSearchCV` and `RandomizedSearchCV` with 5-fold CV. The following models were evaluated:

- **Support Vector Regression (SVR)** with RBF kernel
- **Gaussian Process Regression (GPR)** with RBF + White noise kernel
- **Random Forest Regressor (RF)**
- **Multilayer Perceptron / ANN (MLPRegressor)**

5.2 Height Target — Nonlinear Model Results

Model	R^2_CV	R^2_CV std	R^2_Test	RMSE_CV	RMSE_Test
Random Forest	**0.892**	0.028	**0.904**	136.23 µm	112.95 µm
ANN	0.889	0.065	0.911	130.10 µm	108.70 µm

Model	R^2_CV	R^2_CV std	R^2_Test	RMSE_CV	RMSE_Test
GPR	0.764	0.203	0.891	175.87 µm	120.34 µm
SVR	0.720	0.120	0.893	224.91 µm	119.52 µm

Best model for height: Random Forest (highest and most stable CV R^2 = 0.892, std = 0.028). Its low standard deviation signals **robust generalization** across folds — critical for small-data settings. Optimal hyperparameters: n_estimators=200, max_depth=10, min_samples_split=3, min_samples_leaf=2.

After refitting the Random Forest on the full training data, it achieved **R^2 = 0.9611, RMSE = 84.03 µm**, demonstrating strong in-distribution predictive power.

> **Critical insight for thesis:** The ANN achieved the highest R^2_Test (0.911) but with greater CV variance (std=0.065 vs. 0.028 for RF). In small-data scenarios, **stability of CV performance is more trustworthy** than peak test-set performance on a single split.

5.3 Porosity Target — Nonlinear Model Results

Model	R^2_CV	R^2_CV std	R^2_Test	RMSE_CV	RMSE_Test
SVR	**0.496**	0.157	−2.077	0.339	0.252
ANN	0.415	0.174	−3.202	0.354	0.294
GPR	0.225	0.129	−2.365	0.414	0.263
Random Forest	0.151	0.639	−4.575	0.380	0.339

Despite using nonlinear models, **porosity generalization remains poor** — all models yield strongly negative R^2_Test. The SVR with RBF kernel (C=10, epsilon=0.1, gamma=0.01) achieves the best cross-validated R^2 = 0.496, suggesting it can capture approximately half the variance during training but still fails on the hold-out set. The Random Forest shows catastrophic overfitting (R^2_CV_std = 0.639).

> **Key Learning:** Porosity prediction is fundamentally challenging with only 89 samples and 3 input features. The high CV standard deviation and negative test R^2 suggest that **more data, richer process features (e.g., melt pool monitoring signals, thermal imaging), or physics-informed constraints** are needed to model porosity reliably.

5.4 Aspect Ratio Target

For the **aspect ratio** target, the GPR model was identified as the best nonlinear model and was refitted on the full training dataset, achieving **R^2 = 0.9873, RMSE = 0.0258** — indicating that aspect ratio is highly predictable from the three process parameters.

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6. Comparative Model Summary Across Targets

Target	Best Linear Model (CV R^2)	Best Nonlinear Model	Best NL CV R^2	Best NL Test R^2
Clad Height	OLS / Ridge (~0.76)	Random Forest	0.892	0.904

Target	Best Linear Model (CV R^2)	Best Nonlinear Model	Best NL CV R^2	Best NL Test R^2
Aspect Ratio	Moderate	GPR	High	~0.987
Porosity	Linear Regression (~−0.64)	SVR	0.496	−2.077

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7. Key Findings, Learnings & Thesis Implications

7.1 Small-Data Methodology

The study rigorously applied small-data ML best practices: **5-fold stratified cross-validation as the primary evaluation metric**, pipelines that prevent data leakage (StandardScaler fit only on training folds), and preference for models with low CV variance over those with peak test performance.

7.2 Target-Specific Model Behavior

- **Geometry targets (height, aspect ratio)** are predictable with nonlinear models at $R^2 > 0.89$, suggesting that process parameters govern geometry through learnable, structured relationships.
- **Porosity** exhibits stochastic behavior that is poorly captured by the available features alone, pointing to the need for **expanded feature engineering** (e.g., energy density, substrate temperature, layer interaction effects).

7.3 Multicollinearity as a Recurring Challenge

The high VIF of Power (14.23) and Speed (10.48) creates instability in linear coefficient estimates but does not necessarily hurt tree-based or kernel methods, which are less sensitive to collinearity. This distinction is valuable when selecting model families.

7.4 Residual Non-Normality as a Signal

The strong non-normality of height residuals under OLS (Skewness = 1.874, JB test $p \approx 0$) is a statistically principled indicator that linear regression is insufficient and that the relationship between process parameters and height is heteroscedastic or nonlinear at certain parameter ranges.

7.5 Model Persistence and Deployment

Best models were serialized using `joblib` for future deployment (e.g., `BestNonlinearModelRandomForestheightum.pkl`), laying the groundwork for real-time process monitoring or digital twin integration in an LMD system.

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8. Recommendations for Thesis Development

1. **Feature expansion**: Incorporate energy density (VED/LED) as engineered features with VIF monitoring to balance physical interpretability against collinearity risk.
2. **Data augmentation**: Explore physics-based simulation data to supplement experimental samples, enabling more reliable nonlinear model training.
3. **Uncertainty quantification**: Leverage GPR's native uncertainty estimates for Bayesian optimisation of LMD process parameters.
4. **Transfer learning**: Investigate domain adaptation from related AM processes (LPBF, WAAM) to bolster training signal for porosity prediction.
5. **Ensemble stacking**: Combine RF and ANN base learners under a meta-learner for a potentially more robust height predictor on new parameter regimes.